Supplementary material for PLECTA: Overlap-aware instance segmentation of dense strand networks from binary masks

Oday Allan[a] Igor Ostanin[a]

[a] Multi-Scale Mechanics (MSM), Faculty of Engineering Technology, University of Twente, Enschede, The Netherlands

S1 Instance segmentation: cost function, gates and schedule

This section gives the numerical detail of Section 3.3 of the article. All values are those of the released configuration and were selected on development data only.

S1.1 Skeleton graph

The mask M is binarised at > 0 and skeletonised, and spurs no longer than 3 px are removed iteratively. Skeleton pixels of degree at least three form junction clusters, and adjacent clusters merge into nodes: connectors up to 5 px are absorbed into one node, connectors up to 14 px merge nearby nodes while remaining arms, and free debris up to 2 px is absorbed. Each remaining connected path is stored as an ordered arm with one stub at each end.

S1.2 Stub frames

A stub frame consists of the tip position, the outward tangent and the signed curvature, estimated by a least-squares polynomial fit in arclength measured outward from the tip. The fit is quadratic when the span reaches 18 px and linear otherwise, and a frame is considered reliable only if it rests on at least three samples spanning at least 3 px. Local frames use 24 px of support. Once links form chains, the support extends to 55 px, so that a stub too short to define a direction of its own inherits one from the chain it has joined.

S1.3 Weights and scales of the pairing cost

Junction and gap links use separate weight sets (Table S1). A junction link is scored on direction alone, whereas a gap link is also charged for the distance it spans; junction links therefore set $w_\ell = 0$. The curvature term carries a constant of 30, which brings curvature, in px^{-1}, onto the scale of the angular terms. The parameters ℓ_0 and d_0 set the length scale of the gap penalty and the width of the chord fade, respectively.

S1.4 Penalties, gates and admissibility

Leaving a stub unmatched incurs a penalty p_i rather than being prohibited. Exact maximum-weight partial matching maximises $\sum(p_i + p_j - C_{ij})$ over disjoint pairs, which is equivalent to minimising the total link cost plus p_i for every stub left unmatched. The same penalty determines admissibility, $C_{ij} < p_i + p_j$. This is not an additional threshold: an edge with $C_{ij} \geq p_i + p_j$ makes no positive contribution to the objective and would never be selected by a partial matching.

Table S1. Cost weights and scales of the released configuration. Junction and gap links are scored with separate sets; only the gap set includes a length term.

Symbol	Term	Junction	Gap
w_d	reversal angle θ	0.6	0.2
w_t	chord turns $\varphi_a + \varphi_b$	1.2	1
w_κ	curvature continuity	1	0.5
w_ℓ	length d/ℓ_0	—	0.15
ℓ_0	length scale (px)	—	60
d_0	chord-fade scale (px)	4	3
p_i	price of leaving a stub free	0.62	0.3

Junction stubs carry $p_i = 0.62$, so a junction edge requires $C < 1.24$. Gap stubs carry $p_i = 0.30$, so a gap edge requires $C < 0.60$ and, in addition, $d \leq 85\,\text{px}$, $\theta \leq 0.40\,\text{rad}$ and $\varphi \leq 0.28\,\text{rad}$ at each end. The gap gates are stricter because a gap link asserts continuity across missing, rather than ambiguous, evidence. All values were fixed on development scenes.

S1.5 Round schedule

Rounds $r = 0, \ldots, 8-1$ apply a schedule fraction $\alpha_r = 0.85 + (1 - 0.85)\, r/7$ to the unmatched penalties and to the two gap angle limits, and to nothing else. A lower penalty admits fewer edges and a lower angle limit narrows the gap gates, so the early rounds, in which frame estimates are least reliable, are deliberately conservative; the constraints relax as the assembled chains improve the frame estimates. Only the final round, $\alpha_7 = 1$, runs at the configured parameters, and its result is returned. The complete procedure is:

1. Skeletonise M, prune spurs, merge junction clusters into nodes, and store the remaining paths as arms with a stub at each end.
2. For $r = 0, \ldots, 7$, re-solving both matchings from scratch and carrying only the chain structure forward from round $r - 1$:
   (a) Re-fit every stub frame along the chain to which it now belongs.
   (b) At each node, form candidate edges between stubs of different arms with finite frames and cost, retain those admissible at α_r, and solve one maximum-weight matching per node.
   (c) From the still-unmatched reliable stubs, form gap candidates between different arms and different nodes that pass the distance gate and the α_r-scaled angle gates, and solve one global maximum-weight matching over the admissible candidates.
3. Return the links of round $r = 7$.

S1.6 Joint versus sequential matching

Joint and sequential solutions coincide at simple crossings and diverge only at dense nodes. Over 20 development scenes they never differed at nodes of degree 2–4 ($n = 542$), differed at 22 % of nodes with 5–8 stubs ($n = 187$) and at 69 % of nodes with 9 or more ($n = 233$); at these nodes the joint solution had a lower total cost, by a median of 0.90. This comparison holds the costs of a single node fixed. Within the round schedule, where each matching determines the frames against which the next is scored, an early low-cost choice alters later frame estimates, so the difference is not confined to dense nodes.

S1.7 Instance output

Connected components of accepted arm links define instances. Every arm pixel, and every junction cluster touched by a chain, is assigned to that chain's instance, so intersecting instances share crossing pixels. Isolated one-arm components shorter than 6 px are omitted. Accepted gap links determine identity but are not drawn into the centreline output. All effective parameters, including former code defaults, are stated explicitly in the released JSON configuration.

S2 Depth stage: evidence and solver

This section gives the detail of Section 3.4.2 of the article.

S2.1 Sampling geometry at a crossing

Where two instance centrelines intersect, a short arc of each is sampled through the crossing: the shared *core*, a fixed 7.5 px of arc at every crossing, and the *flanks* on either side of it. Flank samples are kept clear of the other strand, because a sample inside it would measure the superposition of both strands rather than one. A crossing with too few usable flank samples abstains. Earlier versions sized the core from the radii of the two strands; with a fixed core, no radius estimate enters the evidence, and radius errors cannot affect a depth decision. Radii enter only downstream, in the non-interpenetration constraints that determine metric height.

S2.2 Intensity score and noise floor

A single quantity, the median intensity, is read on the core and on the flanks. Let m_i and m_j denote the flank medians of the two strands and m_c the pooled core median. The evidence at the crossing is the score

$$s \;=\; 2\,\frac{|m_c - m_j| - |m_c - m_i|}{\max\!\big(|m_i - m_j|,\; 2\hat{\sigma}\big)}, \tag{S1}$$

whose sign identifies the upper strand, namely the one whose flanks the core resembles more closely, and whose magnitude $|s|$ is the weight of that decision. A crossing abstains when $|s| < 0.40$, a threshold defined directly in score units. The noise estimate $\hat{\sigma}$ is computed per crossing as 1.4826 times the median absolute deviation of the within-strand flank residuals, obtained by centring each strand's flank samples on their own median and pooling the two residual sets. It therefore measures the variation within each strand's appearance, not the difference between the two strands. For compatibility with the output format, each crossing record also contains a probability p; this is a monotone transformation of s, not a calibrated confidence, and is not used in any decision or weighting.

Noise floor in the denominator. By the reverse triangle inequality, the numerator of (S1) is bounded in magnitude by $|m_i - m_j|$, so a denominator equal to that separation drives the score to ± 1 whenever the two strands differ at all. A previous version of the rule bounded the denominator below by a small fixed constant in grey levels, well below the noise level of these images. A maximal score could then arise from a separation smaller than the noise, and because a maximal score exceeds any abstention threshold, such crossings could never abstain. This case was common: a substantial fraction of crossings saturated in this way, and those whose separation was below the noise level were markedly less accurate than those separated by more. Bounding the denominator by $2\hat{\sigma}$ ensures that a high score indicates a separation larger than the noise, and allows the stage to abstain where the evidence is weak.

Removed gradient channel. An earlier version also read the median gradient magnitude along each strand's two edges, at centreline $\pm r\,\mathbf{n}$, on the premise that the upper strand's edges remain sharp through the crossing while those of the lower strand are occluded. On real crossings this channel decided almost no crossings, agreed with the intensity channel no better than chance, and was itself at chance level. Because the channel is not part of the released implementation, these measurements are not reported quantitatively. The cause lies in the sampling geometry: within the core, the $\pm r$ offsets fall inside the other strand, so the measurement is contaminated exactly where it is needed. Removing the channel also removed the gradient image, the two channel weights and the logistic function that combined the channels; with a single channel, s is itself the decision variable.

Comparison with the two-channel rule. The one- and two-channel rules were compared on 20 optically rendered synthetic scenes containing 1342 crossings. The core half-size (7.5 px) was chosen by a sweep on the same set, and the insensitivity of the result to the abstention threshold was also checked on it, so the comparison is in-sample with respect to these two values. Both values were nevertheless inherited rather than newly selected: 7.5 px is the value the previous rule's radius-free fallback already used, and both lie on a plateau. A fixed core performs no worse than oracle per-strand radii anywhere between 6 and 14 px, and accuracy is unchanged over a band of thresholds around 0.40.

At matched coverage, with both rules answering as often as the released rule does at 83.0 %, the one-channel rule matches the accuracy of the two-channel rule, with a difference of $+0.0045$ (95 % CI $[-0.0186, +0.0235]$). When both rules are forced to decide every crossing, including those on which they would abstain, the one-channel rule is less accurate by -0.0142 (CI $[-0.0313, +0.0007]$); neither rule is operated in this mode. The one-channel rule orders its confidence better: the area under the curve of correctness against $|s|$ increases from 0.6531 to 0.7586, a gain of $+0.1056$ (CI $[+0.0659, +0.1425]$). Because $|s|$ is the edge weight of the global ordering described below, better-ordered confidence means that contradictions are resolved at the weakest relations. It also translates into higher accuracy when more abstention is acceptable, by $+0.0054$ at 82.5 % coverage and by $+0.0121$ at 80 %. The one-channel rule therefore matches the accuracy of the two-channel rule, with better-ordered confidence and a simpler evidence path.

S2.3 Global ordering

Each decided crossing contributes a directed edge, oriented according to the sign of s and weighted by $|s|$; p is not used. Within each connected component of the crossing graph, a total order of the instances is derived that maximises the total weight of satisfied relations, and every relation in the component is then re-read from this order. Relations that disagree with the order are reversed rather than discarded, and both the reversal and the original local score are recorded.

Maximising the satisfied weight is the minimum-weight feedback arc set problem. Components of at most 14 instances are solved exactly by dynamic programming over subsets; larger components receive a greedy order refined by adjacent swaps, without an optimality guarantee. Because the order is derived for the whole component, a greedily ordered component can reverse a relation that is not contradicted by any other, which an exact solution never does; a reversal that lies on no cycle only reduces the satisfied weight. When supplied with ground-truth crossings and their true directions, which are acyclic by construction, the procedure still reverses 24 relations across the 20 evaluation scenes, limiting layer agreement to 0.886 instead of 1.000. A variant that orders the condensation of the strongly connected components and reorders only within them reverses none of these relations, but performed worse on the actual, noisy predictions and is not used in the released implementation.

Table S2. Depth-stage entries of the released parameter file. The abstention threshold is in score units, and the core length is fixed, so no strand radius enters the evidence.

Quantity	Value	What it sets
core (px)	7.5	half-length of the shared crossing arc; flat at every crossing, so no per-rod radius enters the evidence path
window (px)	22	arclength sampled either side of a crossing for evidence
min. flank	6	fewer usable flank samples and the crossing abstains
abstain $\|s\|$	0.4	abstention threshold, in score units, not a probability band
exact limit	14	components to this many nodes are ordered exactly, larger ones greedily

S2.4 Layers and metric height

The corrected relations form a directed acyclic graph, whose longest path determines the smallest number of layers consistent with them. Strands that are never in conflict share a layer, so the number of layers follows from the ordering rather than from the number of instances. Each instance is then assigned a height that respects the order and prevents interpenetration at the measured radii, with the stack compacted to the minimum thickness these constraints allow.

S2.5 Configuration

All tunable quantities of both stages, namely the cost weights and gates above, the core and window sizes, the abstention threshold and the size limit of the exact solver, are stored in a single released parameter file, grouped by stage. Table S1 lists the in-plane parameters and Table S2 the depth parameters; every value quoted in this section is taken from that file.

S3 Measures

This section defines every measure reported in the article, together with what it rewards and its principal limitation. None of the measures depends on the rendered width of an instance.

Common-fragment set. All measures below except detection F_1 are computed on a common set of fragments. The mask is skeletonised, spurs of at most 3 px are pruned, skeleton pixels with three or more 8-neighbours are deleted, the remaining 8-connected pieces are labelled, and pieces shorter than 6 px are discarded, so that each fragment is an unambiguous arm between crossings. This partition depends on neither the reference nor any prediction, so every method is evaluated on the same fragments. Each fragment then receives a reference identity and a predicted identity by the same rule. It is assigned to the instance whose mask overlaps the largest number of its pixels, counted on each instance's full mask rather than on a flattened label image so that overlapping instances retain their pixels, provided that this overlap covers at least 30 % of the fragment. Otherwise it is assigned to the most frequent nearest instance label within 6 px, read from a deterministically flattened label image; failing both, it is treated as a singleton cluster on both sides, so that every fragment enters the evaluation. Ties are resolved in favour of the lowest instance identifier and the flattening order is fixed, so the assignment is deterministic. Because junction pixels are removed before labelling, these measures do not register whether a crossing pixel is assigned to one instance or to two, only whether the arms meeting at the crossing are grouped as in the reference.

Sensitivity to the scorer's constants. The four constants above are shared with the method (Section 3.5). The stored predictions of all four methods were therefore re-scored under one-at-a-time perturbations of each constant, on a grid fixed in writing before any result was computed: spur pruning 2–4 px, minimum fragment length 4–8 px, overlap threshold 0.20–0.50 and fallback distance 3–9 px. The perturbations affect scoring only; all stored predictions and configurations predate them. Table S3 reports PLECTA's mean F_1 and its paired margins over each comparator under every perturbation, over the 50 degraded comparison scenes. Across the 10 combinations, PLECTA's mean F_1 ranges from 0.71 to 0.86, and the smallest paired margin over any comparator under any perturbation is 0.11. Only the minimum fragment length changes the level appreciably, and it does so in the same direction for every method, because shorter fragments add difficult, low-evidence units for all of them. The other three constants change the mean by less than 0.002, and no perturbation changes the ranking of the four methods.

Table S3. Sensitivity to the scorer's constants. Stored predictions were re-scored under one-at-a-time perturbations of the four common-fragment constants; Δ columns are paired mean margins, PLECTA minus comparator, over the scenes completed by both methods. The published constants were not changed in response.

Scorer constants	PLECTA F_1	Δ greedy	Δ DNAi	Δ GraFT
published constants	0.82	0.15	0.48	0.34
spur prune 2 px	0.82	0.15	0.48	0.34
spur prune 4 px	0.82	0.15	0.48	0.34
min fragment 4 px	0.71	0.11	0.39	0.27
min fragment 8 px	0.86	0.15	0.51	0.36
overlap threshold 0.20	0.82	0.15	0.48	0.34
overlap threshold 0.40	0.82	0.15	0.48	0.34
overlap threshold 0.50	0.82	0.15	0.48	0.34
fallback distance 3 px	0.82	0.15	0.48	0.34
fallback distance 9 px	0.82	0.15	0.48	0.34

Common-fragment pairwise $\mathbf{F_1}$ is the primary endpoint. A pair of fragments is a true positive when both identities agree, a false positive when the prediction joins fragments of different reference strands, and a false negative when it separates fragments of the same strand:

$$P = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}, \qquad R = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}, \qquad F_1 = \frac{2PR}{P+R}. \tag{S2}$$

It is a clustering measure of the Rand family [1–3], invariant to permutation of the instance labels, and it rewards keeping a strand together across every crossing and gap it passes. Limitations: it is quadratic in fragmentation, so a strand split into n pieces contributes $\binom{n}{2}$ pairs and a single missed join in a fragmented mask is penalised many times; and it weights long strands with many crossings far more heavily than short ones.

Adjusted Rand index [4] counts the same pairs, corrected for chance. The correction matters because, with about a hundred strands per scene, almost every pair belongs to different strands and the uncorrected index is close to 1 for any prediction. Limitation: over the comparison set it never differs from pairwise F_1 by more than 0.02, so it confirms rather than adds information.

Variation of information [5], in bits and lower for better agreement, is the conditional-entropy distance between the two partitions. Its split component, $\mathrm{VI}_{\mathrm{split}} = H(\text{prediction} \mid \text{reference})$, penalises dividing one strand into several, and its merge component, $\mathrm{VI}_{\mathrm{merge}} = H(\text{reference} \mid \text{prediction})$, penalises fusing several strands into one; together they distinguish methods that over-segment from methods that under-segment. Limitation: each component alone is degenerate, since a method that merges everything never splits and scores zero on the split component, so the two are always reported together.

Detection $\mathbf{F_1}$, the measure reported by DNAi, matches predicted to reference *objects* greedily by highest intersection over union, down to 0.1, DNAi's own default threshold, with precision computed over predicted and recall over reference objects. A symmetric placement tolerance of 2 px is applied: a reference pixel counts as found when the prediction passes within this distance, and vice versa, and the intersection is taken as the smaller of the two counts. Without it, two independently traced one-pixel centrelines two pixels apart would not intersect at all, so a strict overlap would measure placement rather than shape, a problem also stated as a design consideration of the thin-structure benchmark FISBe [6]. The real fields show that the tolerance corrects rather than inflates the score. On the reference-derived axes, a skeleton of the annotation itself with no displacement, it changes detection F_1 on B58_110 from 0.93 to 0.93, whereas on the nnU-Net axis of the same field it raises it from 0.18 to 0.60; the tolerance therefore matters only where two readings differ in placement. Because it counts objects, this is the most lenient measure of a fragmented mask and the least discriminative measure reported. Limitation: it is blind to an arm swap, which it scores 1.000 where the pairwise score gives 0.000.

Crossing Fidelity measures junction resolution directly. At every junction node the reference induces a partition of the incident arms, two arms being grouped together exactly when the reference assigns them to the same strand, and the prediction induces another; Crossing Fidelity is the fraction of crossings at which the two partitions are identical. Nodes are the junction clusters of the same pruned skeleton, and the arms of a node are the fragments 8-adjacent to it. Only nodes whose arms carry at least two distinct reference strands are scored, since a node whose arms all belong to one strand is a skeletonisation artefact and scoring it would reward inaction. It is a diagnostic introduced for this study rather than an established metric, although it follows measures that assess correctness at branch points rather than per pixel [7, 8]. Limitations: it is much less discriminative than the pairwise score, so it is used as a count of resolved crossings rather than for ranking; and it carries no information about gaps, so it does not reflect the cost of bridging them. Because it shares its skeleton and assignment with the pairwise score, it provides a finer view of the same evidence rather than independent confirmation, and it is always reported together with its all-unpaired baseline.

Join F_1 counts decisions rather than pairs and is used on the real fields, where the two mask sources fragment differently. A *decision* is a pair of distinct fragments whose endpoints lie within 85 px of each other, the largest gap the method bridges; reference and prediction each either join the pair or not, and precision, recall and their harmonic mean are computed over decisions. Because it is linear rather than quadratic in fragmentation and conditioned on the mask, it poses comparably difficult questions for both mask sources, and splitting and merging appear separately, as low recall and low precision respectively. Limitation: fragment pairs that the mask never brings within reach are not counted, so the measure does not register a strand missing from the mask entirely; it is therefore always reported with the number of decisions each mask poses.

Measures computed but not used as endpoints. GraFT's *filament matched coverage* asks whether each reference strand is matched one-to-one to some detection, and on it PLECTA ranks *below* both comparators: at the highest density of GraFT's own generator, 0.80 against 1.00 and 0.90. It is not used as an endpoint because it measures existence rather than correctness. A method that outputs more objects than the scene contains has a candidate for every strand, each method's matched coverage equals the smaller of one and its output-to-reference object ratio to within 0.16, and requiring each detection to cover half of the axis it claims restores the ranking. *Centreline Dice* [9] was evaluated as an alternative matching criterion and not adopted. It tests each side's skeleton against the other's *volume*, which a centreline grouping does not have, so it reduces to the placement test that the tolerance above is designed to correct. No clDice results are reported.

Validity of a reference. A defective reference cannot be detected by any of the measures above: it lowers every method's score slightly without changing their ranking. A reference is therefore accepted only if, scored as a prediction of itself, it obtains a perfect score on every measure. Any shortfall indicates that the reference disagrees with its own fragment assignment and bounds the score any method could obtain against it.

S3.1 Statistical analysis

Synthetic-scene means and density strata are reported with 95 % nonparametric bootstrap intervals (20,000 replicates; seed 20,260,810), resampled with stratification by density; the comparison between clean and degraded masks resamples paired scene differences within density. Per-coverage points in the comparison figures are plain means of ten scenes, without intervals. On the real SEM fields the inferential unit is the field, and the 2566 reference crossings entering the crossing measures are a count pooled over six condition-specific evaluations of the three fields, not a sample size. Component ablations are one-at-a-time experiments on all 84 development scenes and do not use the evaluation set.

S4 Comparator configuration

Section 4.5 of the article summarises how each comparator was adapted and at what cost. This section describes the mechanism behind each adaptation, so that the comparison can be verified.

DNAi clusters junction pixels by single linkage and erases a disk at each cluster. At its released linkage distance of 10 px, nearby crossings chain into one cluster, which at 60 % coverage reaches 89 px across and is replaced by a single small erasure disk, leaving a whole tangle as one instance. The released default therefore does not reflect the method's capability on these scenes. Reducing the distance to 2 px on development scenes raised development F_1 and reduced the merge component of the variation of information by about a factor of three, confirming this mechanism; the tuned configuration is the one reported. Two design choices account for the remaining difference. DNAi commits to a pairing at every junction of degree two to four without a cost threshold, so it cannot decline a continuation in the way a penalised unmatched option allows; and it attempts no junction above degree four, resolving 0.02 of those crossings. These are about a tenth of all crossings, but they carry 37.2 % of same-strand fragment pairs on the degraded masks, since a single unresolved junction separates every pair that spans it.

GraFT's errors are predominantly splits rather than merges (split component of the variation of information 1.17, merge component 0.39). This follows from its greedy longest-path cover, which cannot revise an early decision once the scene forms a single connected graph, as it does above 30 % coverage.

SIFNE enhances the image with an orientation filter transform, thresholds it by Otsu's method and thins it, and thereafter operates on the binary skeleton alone. It marks every pixel whose eight-connected neighbourhood sum reaches three as a crossing, erases each such pixel together with a square neighbourhood, and pairs the resulting fragment termini within an angular search fan, bidirectionally and one pairing at a time [10]. Since every step after thinning operates on a binary axis, the mask is supplied as the thinned array, and all subsequent stages run as released. Only the graphical callbacks were re-implemented, to allow unattended execution, and the manual-correction stage was not used, since no comparator here involves manual intervention.

One released default is degenerate at these densities and was re-selected on development scenes. The erasure neighbourhood is 7 pixels square and is applied at every pixel whose eight-connected neighbourhood sum reaches three, so the erased area grows with the number of crossings: on scenes at 20, 40 and 60 % coverage, the

remaining mask is 51, 44 and 31 % of the input, and the evidence is removed before any pairing is attempted. On the sparse single-molecule images for which the method was designed, the erased fraction is small. Selection proceeded in two stages, with both grids fixed in advance: first the three settings that determine how much of the mask is retained, then the linkage gates at the best setting of the first stage. Only the ten development scenes were used. The selected configuration reduces the erasure to 5 pixels square and the search radius to 35 pixels, raising development F_1 from 0.537 to 0.749 and Crossing Fidelity from 0.502, below the all-unpaired baseline, to 0.831. On the comparison set, the same change raises F_1 from 0.55 to 0.71; both values are reported.

Inspection of the source code shows that the search-fan angle has no effect at the selected angular gate, because for any fan wider than sixty degrees the gate is the stricter of the two tests. The neighbourhood radius used by the orientation filter, by contrast, remains effective when the mask is supplied directly, because the same value sets the arclength over which every tip direction is estimated, which is the only evidence its pairing uses. SIFNE's crossing excision does not bias the scoring: although the deleted crossing pixels are never restored, the common-fragment scorer also removes junction pixels and evaluates only whether the arms meeting there are grouped as in the reference, and SIFNE links across the excised crossing, so it makes exactly the decision that the endpoint measures.

Basu et al. has no released implementation: the only code published by its authors accompanies their earlier centreline-localisation paper, and its address no longer resolves, with no archived copy available. The method is therefore reimplemented from its published description and reported as a reimplementation baseline rather than as an external result. Only its geometric part is reimplemented: a minimum spanning tree over the centreline pixels, removal of every bifurcation node, and recombination of two- and three-segment groups scored by the curvature energy of a fitted smoothing cubic spline [11] minus a bond energy, selected by a set-partitioning integer program that is re-solved until the segment set no longer changes. This part uses geometry alone and therefore accepts an axis mask directly. The intensity-based part, a filter bank followed by a reverse diffusion that moves particles onto intensity ridges, is not reimplemented, because evaluating a substitute front end would measure that substitute rather than the published method; this baseline therefore evaluates the recombination step only.

Its two user parameters, the maximum linking distance and the maximum tangent mismatch, were selected on the same ten development scenes, by the same criterion as for the other comparators. The values published for the original simulated benchmark are replaced by 16 and 10 here; with the published pair, the reimplementation scores 0.076 on these masks, because a linking distance of two pixels permits almost no recombination when the gaps in the mask are wider. Both results are recorded. The original description leaves two further quantities unspecified that affect the result, the smoothing of the fitted spline and the quadrature of the bond energy; each was set in the direction favourable to the method, and the choice is recorded. On Crossing Fidelity the recombination decides junctions almost as well as the exact matching does, but it loses the pairwise margin in the assembly built from those decisions, because a pairwise score over fragments is quadratic in fragmentation, whereas a junction count is not.

S4.1 Paired margins

Paired against PLECTA over the degraded set, the Crossing Fidelity margins are +0.065 [+0.047, +0.083] over the greedy baseline, +0.123 [+0.085, +0.165] over GraFT and +0.185 [+0.158, +0.211] over DNAi. On clean axes PLECTA resolves 0.83 of crossings. Leaving every arm unpaired resolves 0.09 of them; this is the all-unpaired baseline against which the rate is read, a deterministic policy rather than a chance level. Crossing Fidelity is much less discriminative than the pairwise score: at 60 % coverage it spans 0.26 between the best and worst methods, whereas pairwise F_1 spans 0.54. Resolution rates decline for every method at the highest-degree nodes: pooled over the 3638 reference crossings of the degraded set, PLECTA resolves 0.81 exactly, but only 0.41 above degree four, where no comparator is higher.

A restricted comparison with DNAi scores only the evidence that a method limited to junctions of degree four or less, and unable to bridge gaps, can reach. It changes the clean-axis difference from +0.414 to +0.284 [+0.262, +0.307], still in favour of PLECTA in 49 of 50 scenes.

S5 Depth measures

The measures of Section S3 assess which centreline fragments belong to the same instance and are insensitive to depth: two reconstructions that agree on every instance but disagree on which strand lies on top at every crossing receive identical scores. The depth stage is therefore evaluated with the following measures.

Crossing-order accuracy. Each reference crossing specifies which instance lies on top; the reconstruction either names one or abstains. Three quantities are reported, never in isolation: *decided only*, the fraction of committed decisions that are correct, which alone is not informative because abstaining on every difficult crossing yields 1.000; *all crossings*, which counts an abstention as an error and is the primary quantity; and the *abstention*

rate. The decision is binary and symmetric, so chance is exactly 0.5 and only the margin above it is informative. A predicted crossing is credited only if it involves the same instance pair as a reference crossing and lies within the placement tolerance, so spurious crossings cannot improve the score.

Depth order over instance pairs. This global counterpart asks, for a pair of instances, whether the reconstruction places the deeper one deeper. It is reported over two sets. Over *crossing pairs*, it measures the stack where evidence exists. Over *all pairs*, it also includes instances that never cross, whose relative depth is set by the solver's tie-breaking rule rather than by the image. The two are never pooled, and the fraction of pairs that share a connected component is reported alongside them, so that the second can be interpreted. Both are invariant to monotone rescaling of depth and therefore describe the arrangement rather than its units. Kendall's τ over the same pairs equals $2a - 1$ for accuracy a and is not reported separately.

Layer structure. Layers are an ordinal quotient of depth: a reconstruction can order every pair correctly but divide the film into a different number of layers. Layer agreement, the fraction of instances placed in the reference layer, is therefore reported separately.

S6 Development and evaluation split

All parameters were fixed on a separate development set and never on the scenes reported in the article. Development used 84 scenes spanning 20–60 % coverage. All segmentation parameters, gates, penalties and the round schedule were set on these scenes, as was each comparator's configuration, so that no method was tuned on the scenes on which it is scored and all were tuned on the same scenes. The evaluation set comprises 128 scenes, 32 at each of 20, 30, 40 and 60 % coverage. The generator seeds of both sets are listed in the seed manifest released with the code.

The synthetic intervals describe sampling variability within this generator; the ranking is additionally tested on an independently published generator and on real SEM masks. The generator's shape parameters were also varied. Changing its bending stiffness from 50 to 150 % of the default, which moves the median centreline radius from 975 to 308 px, changed the level means of F_1 by at most 0.05 over 4 series, compared with a within-level scene-to-scene standard deviation of 0.05. Changing the curve wavelength from 10 to 40 px changed them by at most 0.08; the one series of 4 whose variation exceeded its own noise is non-monotonic, with its minimum at the default, and no mechanism for this was identified. Both variations were measured on development scenes, 10 scenes per level, and are reported as plain means.

S7 Manual annotation procedure

Reference annotations were created with an interactive instance-segmentation tool for greyscale SEM images of crossing strands, developed in-house. Each bundle is stored as a separate mask, so that crossing pixels can belong to several strands, and the saved annotation file is read directly by the evaluation, without intermediate export or conversion.

Two optional forms of assistance are available during tracing. Strokes can snap to the local intensity ridge, with polarity calibrated from the existing annotation, and can be re-rendered as a smoothing spline whose centreline is shifted to the measured intensity crest only where that crest is consistent. Apparent widths are measured from perpendicular full-width-at-half-maximum profiles, referenced to the local minima on either side of the peak; the reported width is twice the smaller half-width, which limits overestimation when a crossing strand merges into the profile. These widths are not used by the centreline-based segmentation scores. The tool proposes gap bridges between collinear endpoints of different strands, but each proposal must be accepted by the annotator, so strand identity through crossings and gaps remains a manual decision.

References

[1] William M. Rand. Objective criteria for the evaluation of clustering methods. *Journal of the American Statistical Association*, 66(336):846–850, 1971. doi:10.1080/01621459.1971.10482356.

[2] Enrique Amigó, Julio Gonzalo, Javier Artiles, and Felisa Verdejo. A comparison of extrinsic clustering evaluation metrics based on formal constraints. *Information Retrieval*, 12(4):461–486, 2009. doi:10.1007/s10791-008-9066-8.

[3] Ignacio Arganda-Carreras, Srinivas C. Turaga, Daniel R. Berger, Dan Cireşan, Alessandro Giusti, Luca M. Gambardella, Jürgen Schmidhuber, Dmitry Laptev, Sarvesh Dwivedi, Joachim M. Buhmann, Ting Liu, Mojtaba Seyedhosseini, Tolga Tasdizen, Lee Kamentsky, Radim Burget, Václav Uher, Xiao Tan, Changming Sun, Tuan D. Pham, Erhan Bas, Mustafa G. Uzunbas, Albert Cardona, Johannes Schindelin, and H. Sebastian Seung. Crowdsourcing the creation of image segmentation algorithms for connectomics. *Frontiers in Neuroanatomy*, 9:142, 2015. doi:10.3389/fnana.2015.00142.

[4] Lawrence Hubert and Phipps Arabie. Comparing partitions. *Journal of Classification*, 2(1):193–218, 1985. doi:10.1007/BF01908075.

[5] Marina Meilă. Comparing clusterings—an information based distance. *Journal of Multivariate Analysis*, 98(5):873–895, 2007. doi:10.1016/j.jmva.2006.11.013.

[6] Lisa Mais, Peter Hirsch, Claire Managan, Ramya Kandarpa, Josef Lorenz Rumberger, Annika Reinke, Lena Maier-Hein, Gudrun Ihrke, and Dagmar Kainmueller. FISBe: A real-world benchmark dataset for instance segmentation of long-range thin filamentous structures. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 22249–22259, 2024. doi:10.1109/CVPR52733.2024.02100.

[7] Todd A. Gillette, Kerry M. Brown, and Giorgio A. Ascoli. The DIADEM metric: Comparing multiple reconstructions of the same neuron. *Neuroinformatics*, 9(2-3):233–245, 2011. doi:10.1007/s12021-011-9117-y.

[8] Elizabeth P. Reilly, Jeffrey S. Garretson, William R. Gray Roncal, Dean M. Kleissas, Brock A. Wester, Mark A. Chevillet, and Matthew J. Roos. Neural reconstruction integrity: A metric for assessing the connectivity accuracy of reconstructed neural networks. *Frontiers in Neuroinformatics*, 12:74, 2018. doi:10.3389/fninf.2018.00074.

[9] Suprosanna Shit, Johannes C. Paetzold, Anjany Sekuboyina, Ivan Ezhov, Alexander Unger, Andrey Zhylka, Josien P. W. Pluim, Ulrich Bauer, and Bjoern H. Menze. clDice — a novel topology-preserving loss function for tubular structure segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 16560–16569, 2021. doi:10.1109/CVPR46437.2021.01629.

[10] Zhen Zhang, Yukako Nishimura, and Pakorn Kanchanawong. Extracting microtubule networks from superresolution single-molecule localization microscopy data. *Molecular Biology of the Cell*, 28(2):333–345, 2017. doi:10.1091/mbc.E16-06-0421.

[11] Paul Dierckx. *Curve and Surface Fitting with Splines*. Monographs on Numerical Analysis. Oxford University Press, Oxford, 1993. ISBN 978-0-19-853441-9.